

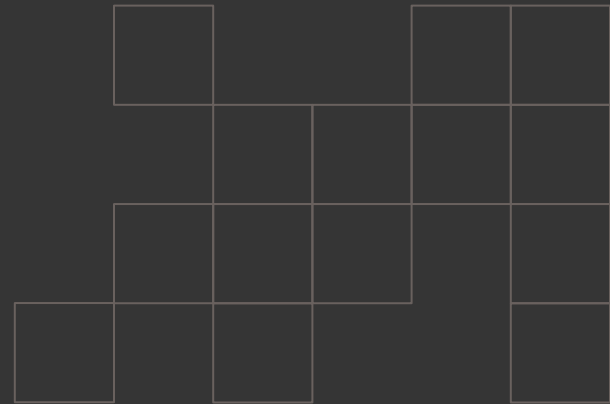
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Customer Satisfaction & Sentiment Analysis

By Akmal Zuhdy Prasetya
Bootcamp Data Analysis & Business Intelligence Batch 19

[Dataset, Notebook, & PBI File](#)

Data Understanding & Preparation



Understanding Dataset & Structure

Note

Dataset terdiri dari 1,462 data survey pelanggan. Terdapat 10 variabel utama, mencakup:

- Rating kepuasan pelanggan
- Feedback teks (ulasan pelanggan)
- Dimensi layanan (customer service, fitur, dll)

Data mencerminkan pengalaman pelanggan terhadap ticketing system

```
RangeIndex: 1462 entries, 0 to 1461
```

```
Data columns (total 10 columns):
```

#	Column	Non-Null Count	Dtype
0	id_survey	1462 non-null	object
1	date_of_survey	1462 non-null	object
2	ticket_system	1462 non-null	object
3	overall_rating	787 non-null	float64
4	customer_service	787 non-null	float64
5	features	787 non-null	float64
6	value_for_money	787 non-null	float64
7	ease_of_use	787 non-null	float64
8	likelihood_to_recommend	787 non-null	float64
9	overall_text	787 non-null	object

```
dtypes: float64(6), object(4)
```

Data Quality Assessment

Note

Ditemukan missing values signifikan (~46%) pada kolom:

- rating
- atribut layanan
- ulasan teks

Hanya 787 data (~54%) yang memiliki informasi lengkap

Insight

Tidak semua pelanggan mengisi survey secara lengkap

Implication

Analisis difokuskan pada data yang lengkap dan valid

```
RangeIndex: 1462 entries, 0 to 1461
```

```
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#	Column	Non-Null Count	Dtype
---	-----	-----	-----
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8	likelihood_to_recommend	787 non-null	float64
9	overall_text	787 non-null	object

```
dtypes: float64(6), object(4)
```

Data Cleaning & Preparation

Note

Proses ini:

- Menghapus data dengan missing values pada variabel utama
- Menghasilkan dataset bersih sebanyak 787 data
- Mengubah format tanggal menjadi datetime

Index: 787 entries, 1 to 1461

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	id_survey	787 non-null	object
1	date_of_survey	787 non-null	datetime64[ns]
2	ticket_system	787 non-null	object
3	overall_rating	787 non-null	float64
4	customer_service	787 non-null	float64
5	features	787 non-null	float64
6	value_for_money	787 non-null	float64
7	ease_of_use	787 non-null	float64
8	likelihood_to_recommend	787 non-null	float64
9	overall_text	787 non-null	object

dtypes: datetime64[ns](1), float64(6), object(3)

Key Variables Identification

Customer Satisfaction Metrics

Overall Rating → **CSAT**

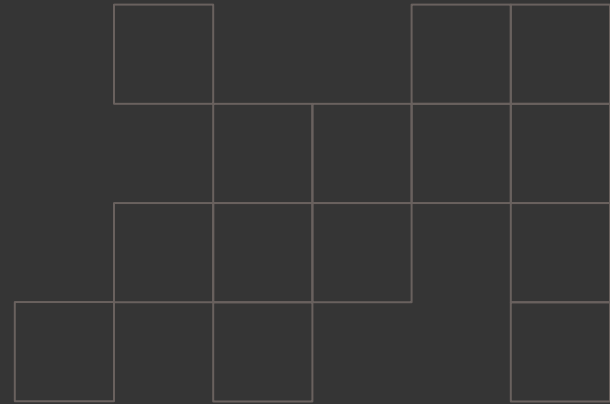
Likelihood to Recommend →
NPS

Ease of Use → **CES**

Sentiment Analysis

Overall Text → **Sentiment Analysis**

Method Analysis



Customer Satisfaction Score (CSAT)

Note

CSAT dihitung berdasarkan proporsi pelanggan yang puas, dengan kriteria:

- Rating $\geq 4 \rightarrow$ satisfied
- Rating $< 4 \rightarrow$ not satisfied

CSAT Score

Measure the customer's overall satisfaction score (CSAT) with the following formula:

$$CSAT = \frac{\Sigma \text{total satisfaction score}}{\text{number of responded customer} \times \text{max rating}}$$

The max rating is inserted to convert the CSAT score into percentage.

CSAT score can be classified into categories based on the result. There is no absolute threshold for each categories but the following is the common threshold:

- $\geq 90\%$: Excellent
- 75%-90%: Good
- 60-75%: Fair
- $<60\%$: Poor

Net Promoter Score (NPS)

Note

NPS menghitung kemungkinan pelanggan merekomendasikan produk, dengan kriteria:

- 0-6 → Detractor
- 7-8 → Passive
- 9-10 → Promoter

NPS Score

To calculate the NPS score, first we must convert the **would_you_recommend** column into proper NPS Category based on the rating value:

- Promoter: Rating 9-10
- Passive: Rating 7-8
- Detractor: Rating < 7

Calculate the NPS Score with the following formula

$$NPS = \frac{Promoter - Detractor}{Total Survey Responded}$$

Customer Effort Score (CES)

Note

CES mengukur kemudahan penggunaan sistem menggunakan variabel ease_of_use sebagai proxy.

CES Score

Measure CES with the following formula

$$CES = \frac{\Sigma \text{total effort score}}{\text{number of responded customer} \times \text{max rating}}$$

The background is a dark gray. In the top right corner, there is a large, abstract shape with a rainbow gradient (red, orange, yellow, green, blue) that tapers to a point. To the left of this shape is a dark gray, rounded, cone-like shape. In the bottom right corner, there is a grid of white-outlined squares, some of which are missing, creating a fragmented pattern.

Sentiment Analysis

Sentiment Analysis Method

Note

Analisis dilakukan pada data ulasan pelanggan (overall_text), dengan tahapan:

- Text preprocessing (lowercasing, menghapus tanda baca dan stopwords)
- Sentiment classification (menggunakan pendekatan lexicon-based / model sederhana)
- Labeling hasil (positive, neutral, negative)

Text Cleansing

In order to get more accurate sentiment, several text cleansing need to be done. However, in most of recent sentiment analysis models and algorithm, the only text cleansing needed are as follows:

- Clean double whitespace
- Clean URL/website
- Clean username (mostly in social media or digital text)

Text Preprocessing

Note

Membersihkan teks menggunakan fungsi `cleansing_text()`, dimana prosesnya:

- Menghapus spasi berlebih
- Mengganti URL menjadi placeholder (`http`)
- Mengganti username menjadi (`@user`)

Proses ini menstandarkan teks sebelum analisis sentimen

index	overall_text	clean_text
1	Customer tickets managements	Customer tickets managements
2	After 6 months of using the Zoho desk, we shifted to different software, but my experience using Zoho was great! Indeed, I would still recommend using Zoho after all.	After 6 months of using the Zoho desk, we shifted to different software, but my experience using Zoho was great! Indeed, I would still recommend using Zoho after all.
4	Pros:Zendesk has always been one of the go-to solutions for helpdesk software, but they've really streamlined their system over the last few years. It's simple and easy to use with straightforward options for even those who are new to using a helpdesk system.Zendesk is basically an email support system, where all emails sent will be routed to your Zendesk dashboard and show up as support tickets. Tickets are tagged with its own ID and even cross-referenced to see if a particular sender has submitted any tickets prior, so it's easy to check through the support history. There are also pre-defined responses, allowing for quick and easy replies for typical queries.Of course, these are actually what you might believe every helpdesk system should provide, but you'll be surprised at how much you have to pay and that some don't even have these options at all. Zendesk is actually very affordable and is easily scalable.	Pros:Zendesk has always been one of the go-to solutions for helpdesk software, but they've really streamlined their system over the last few years. It's simple and easy to use with straightforward options for even those who are new to using a helpdesk system.Zendesk is basically an email support system, where all emails sent will be routed to your Zendesk dashboard and show up as support tickets. Tickets are tagged with its own ID and even cross-referenced to see if a particular sender has submitted any tickets prior, so it's easy to check through the support history. There are also pre-defined responses, allowing for quick and easy replies for typical queries.Of course, these are actually what you might believe every helpdesk system should provide, but you'll be surprised at how much you have to pay and that some don't even have these options at all. Zendesk is actually very affordable and is easily scalable.
6	It has been very useful so far to integrate multiple brands but still allowing us to use the same customer service team.	It has been very useful so far to integrate multiple brands but still allowing us to use the same customer service team.
8	Pros:It's easy to use and very intuitive.We have had very few issues with the functionality and requires minimal training for both customers and agents.	Pros:It's easy to use and very intuitive.We have had very few issues with the functionality and requires minimal training for both customers and agents.
9	Pros:We have connected Zendesk with a lot of our other applications.	Pros:We have connected Zendesk with a lot of our other applications.
12	Its best, but need extra improvement from UI Side	Its best, but need extra improvement from UI Side
15	Served us incredibly well for years, the macros and auto-responders cut down a lot of work for our CS team. It allowed us to add a cost-effective, personalised voice mail, call recording and a comprehensive help desk to support our customers.	Served us incredibly well for years, the macros and auto-responders cut down a lot of work for our CS team. It allowed us to add a cost-effective, personalised voice mail, call recording and a comprehensive help desk to support our customers.
16	I have had a fantastic experience with both the software and customer service. Bang for buck is right on. Great UI/UX all around and make sure we are able to receive feedback through various channels and integrate with Jira.	I have had a fantastic experience with both the software and customer service. Bang for buck is right on. Great UI/UX all around and make sure we are able to receive feedback through various channels and integrate with Jira.
19	Freshdesk has enabled us to manage customer tickets in a disinctive way, and this has helped us improve the speed of response and also to not leave any outstanding tickets within the records, in addition to that it helped a lot by improving quality control tools as well as measuring customer satisfaction.	Freshdesk has enabled us to manage customer tickets in a disinctive way, and this has helped us improve the speed of response and also to not leave any outstanding tickets within the records, in addition to that it helped a lot by improving quality control tools as well as measuring customer satisfaction.
22	Knowledgebase, tickets, satisfied users.	Knowledgebase, tickets, satisfied users.
23	Zoho is 9 and plus rated in comparison to other platforms in the market and it's the best value on investment.	Zoho is 9 and plus rated in comparison to other platforms in the market and it's the best value on investment.
24	Pros:very easy to use and fast interface that helps to manage our business	Pros:very easy to use and fast interface that helps to manage our business
25	The interface is simple, intuitive, and manageable. It is an excellent product for organizing and handling requests and cases.	The interface is simple, intuitive, and manageable. It is an excellent product for organizing and handling requests and cases.
26	Customization, integrations, workflows, SLAs	Customization, integrations, workflows, SLAs
27	We've been using Zendesk for years, it has made email communication/ task management extremely easy and intuitive. All of out team love it	We've been using Zendesk for years, it has made email communication/ task management extremely easy and intuitive. All of out team love it
30	Perfect, alot of feature are with benefit and incentives	Perfect, alot of feature are with benefit and incentives
34	Overall experience with Zoho Desk that they will be there for you support 24/7	Overall experience with Zoho Desk that they will be there for you support 24/7
35	We went from a very clunky process to using Freshdesk, which revolutionised our response time to customers.	We went from a very clunky process to using Freshdesk, which revolutionised our response time to customers.
39	Overall FreshDesk is great for our needs. We are a smaller company so meneuvering the tickets isn't overwhelming and the website provides helpful tools that save tons of time.	Overall FreshDesk is great for our needs. We are a smaller company so meneuvering the tickets isn't overwhelming and the website provides helpful tools that save tons of time.

Sentiment Scoring

Note

Proses scoring::

- Menggunakan library TextBlob
- Setiap teks diberi skor polarity, dengan range -1 (negatif) sampai +1 (positif)

Dimana hasil berupa nilai numerik untuk setiap ulasan

```
# Number of Sentiment
def get_sentiment_score(text):
    return TextBlob(text).sentiment.polarity

df_clean['sentiment_score'] = df_clean['clean_text'].apply(get_sentiment_score)
df_clean[['clean_text', 'sentiment_score']].head(10)
```

	clean_text	sentiment_score
1	Customer tickets managements	0.00
2	After 6 months of using the Zoho desk, we shif...	0.50
4	Pros:Zendesk has always been one of the go-to ...	0.19
6	It has been very useful so far to integrate mu...	0.12
8	Pros:It's easy to use and very intuitive.We ha...	0.07
9	Pros:We have connected Zendesk with a lot of o...	-0.12
12	Its best, but need extra improvement from UI Side	0.50
15	Served us incredibly well for years, the macro...	0.37
16	I have had a fantastic experience with both th...	0.41
19	Freshdesk has enabled us to manage customer ti...	0.50

Sentiment Classification

Note

Skor sentimen dikonversi menjadi kategori:

- score > 0 → Positive
- score = 0 → Neutral
- score < 0 → Negative

Maka setiap ulasan memiliki label sentimen

```
# klasifikasi
def classify_sentiment(score):
    if score > 0:
        return 'Positive'
    elif score == 0:
        return 'Neutral'
    else:
        return 'Negative'
```

```
df_clean['sentiment_label'] = df_clean['sentiment_score'].apply(classify_sentiment)
df_clean['sentiment_label'].value_counts()
```

	count
sentiment_label	
Positive	636
Neutral	118
Negative	33

Sentiment Aggregation

Note

Menghitung jumlah dan persentase tiap kategori sentimen. Digunakan untuk:

- Melihat distribusi persepsi pelanggan
- Mendukung analisis kepuasan pelanggan

```
# distribusi sentimen (%)  
sentiment_dist = df_clean['sentiment_label'].value_counts(normalize=True) * 100  
sentiment_dist
```

	proportion
sentiment_label	
Positive	80.81
Neutral	14.99
Negative	4.19

Validation with Rating Data

Note

Membandingkan hasil sentimen dengan rating pelanggan.

Proses menggunakan crosstab antara sentiment dan rating.

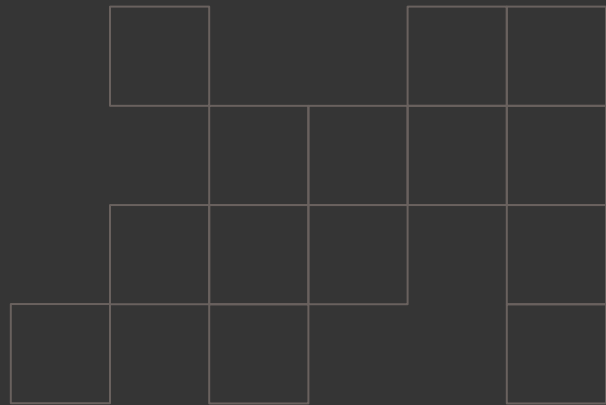
Bertujuan untuk memastikan konsistensi antara skor numerik (rating) dan persepsi tekstual (review).

```
# crosscheck
```

```
pd.crosstab(df_clean['sentiment_label'], df_clean['overall_rating'])
```

1 to 3 of 3 entries Filter ?					
sentiment_label	1.0	2.0	3.0	4.0	5.0
Negative	0	1	6	10	16
Neutral	2	0	7	36	73
Positive	0	3	25	205	403

Dashboard

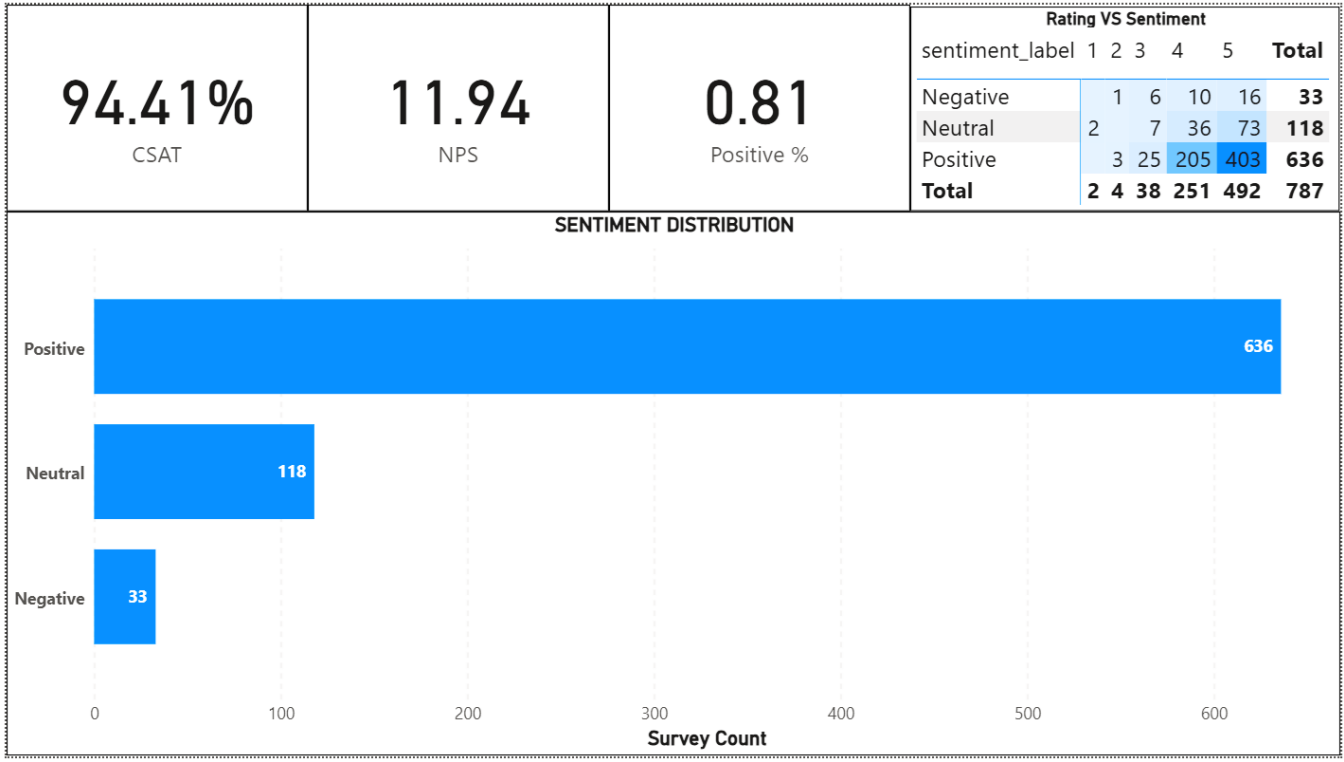


Customer Satisfaction & Sentiment Analysis

CSAT, NPS, CES & Sentiment Insights

Key Findings

- CSAT tinggi (94.41%) → **Mayoritas pelanggan puas**
- NPS relatif rendah (11.94) → **Kepuasan tinggi, tapi loyalitas belum optimal**
- Sentiment positif dominan (~81%) → **Persepsi pelanggan sangat baik**
- Terdapat mismatch dimana beberapa rating tinggi (5) memiliki sentiment negatif → **Indikasi hidden dissatisfaction**

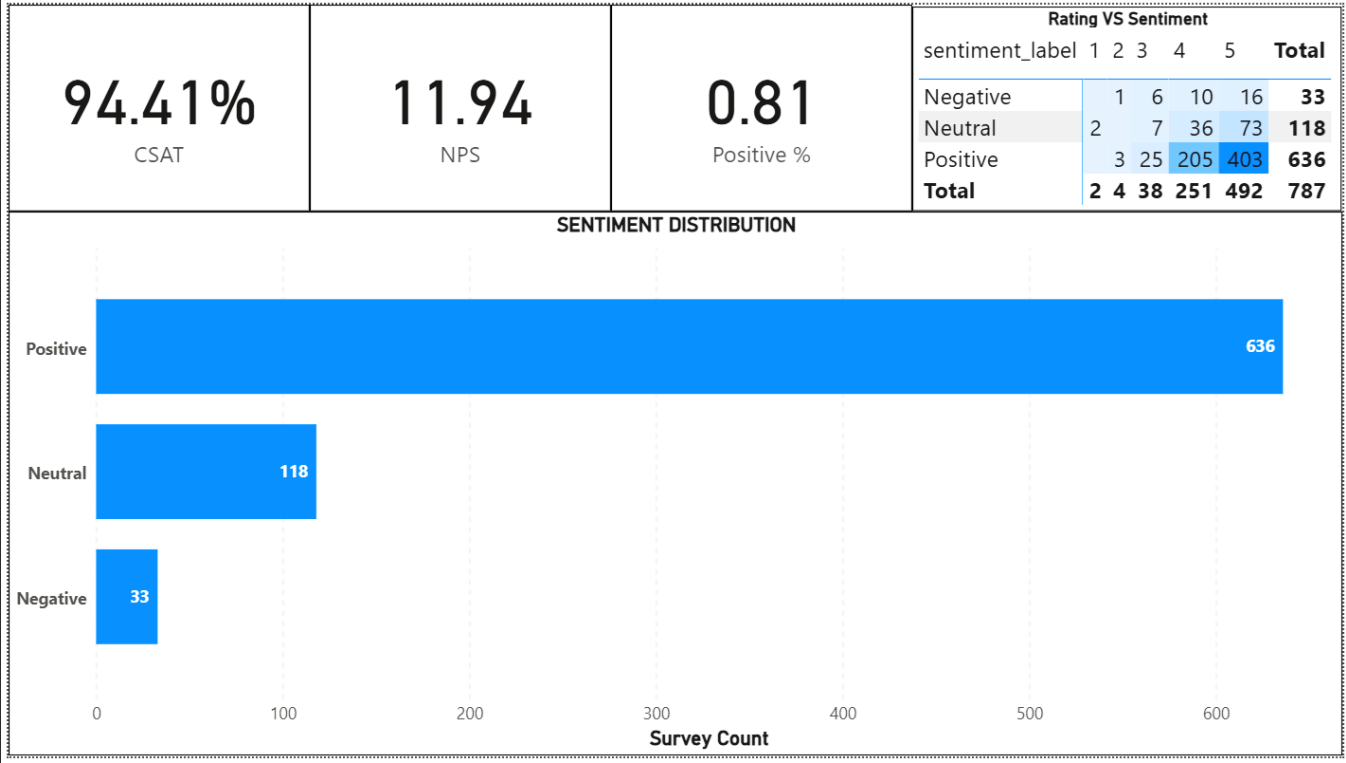


Customer Satisfaction & Sentiment Analysis

CSAT, NPS, CES & Sentiment Insights

Business Insights

- Pelanggan cenderung puas, tetapi belum cukup terdorong untuk merekomendasikan
- Neutral sentiment cukup signifikan, ada peluang peningkatan experience
- Ulasan negatif meskipun kecil tetap kritikal, berpotensi berdampak ke churn

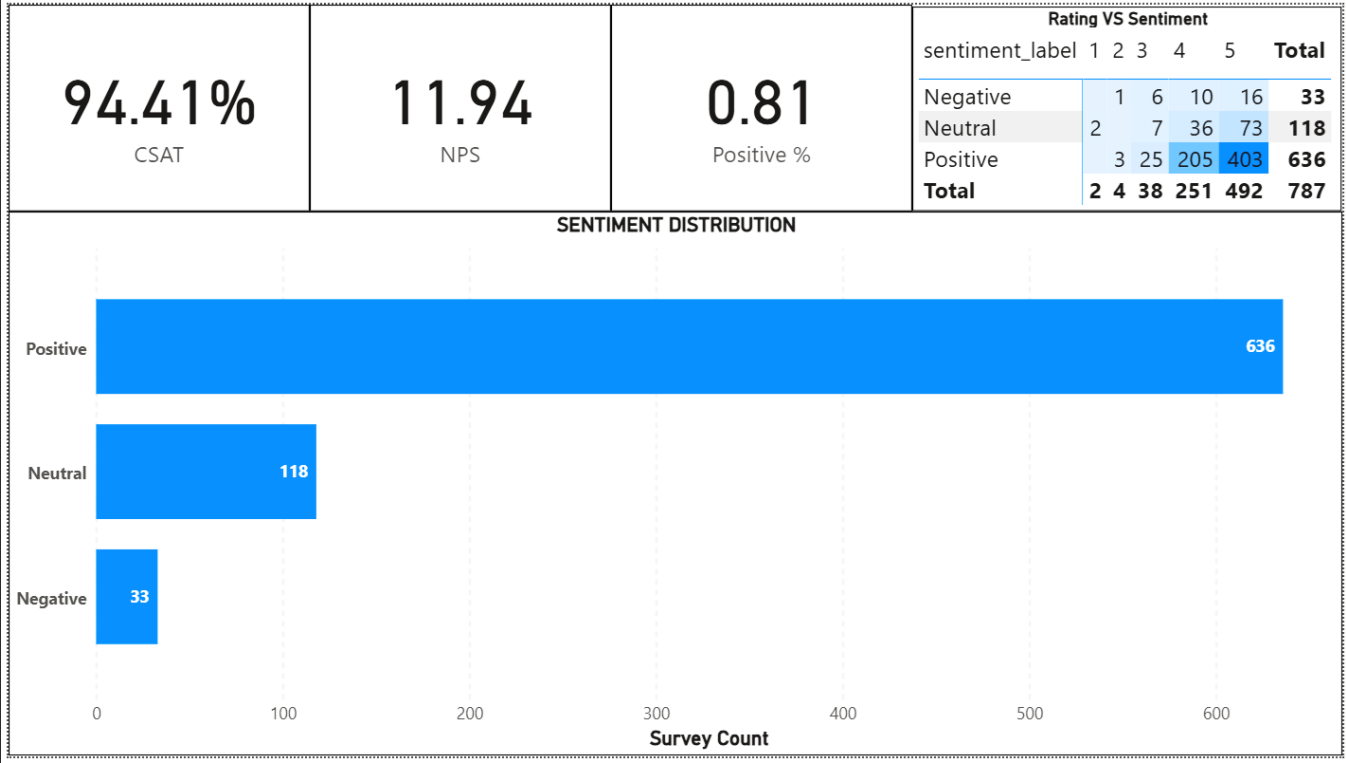


Customer Satisfaction & Sentiment Analysis

CSAT, NPS, CES & Sentiment Insights

Recommendations

- **Improve Customer Experience (CES)**, dengan menyederhanakan proses pelayanan
- **Convert Passive to Promoter**, dengan fokus ke customer dengan rating 4
- **Address Negative Feedback**, analisis keyword utama dari review negatif
- **Leverage Positive Feedback**, gunakan untuk marketing & testimonial



REFLECTION QUESTION 1

Q

Mengapa penting bagi perusahaan untuk menganalisis kepuasan pelanggan dan sentimen dari ulasan yang mereka berikan? Jelaskan bagaimana insight dari kedua analisis tersebut dapat membantu meningkatkan kualitas layanan atau pengalaman pelanggan.

A

Analisis CSAT, NPS, dan CES memberikan gambaran kuantitatif tentang kepuasan dan loyalitas pelanggan, sedangkan sentiment analysis membantu memahami persepsi pelanggan dari ulasan mereka.

Dengan menggabungkan keduanya, perusahaan bisa mendapatkan insight yang lebih lengkap, termasuk menemukan ketidaksesuaian seperti rating tinggi tetapi sentimen negatif. Insight ini dapat digunakan untuk mengidentifikasi masalah dan meningkatkan kualitas layanan serta pengalaman pelanggan.

REFLECTION QUESTION 2

Q

Data understanding dan preprocessing penting banget karena ngaruh langsung ke hasil analisis. Kalau data nggak dipahami atau masih kotor (misalnya ada URL, typo, dll), hasil sentiment bisa melenceng. Dengan data yang sudah bersih dan rapi, analisis jadi lebih akurat dan insight yang dihasilkan lebih bisa dipercaya.

A

Data understanding dan preprocessing sangat berpengaruh terhadap akurasi analisis karena menentukan kualitas data yang digunakan. Jika data tidak dipahami dengan baik atau masih mengandung noise, hasil analisis bisa menjadi tidak akurat.

Dengan melakukan pembersihan dan persiapan data terlebih dahulu, hasil analisis menjadi lebih valid dan dapat diandalkan untuk pengambilan keputusan.



Thank you